

DATA STRUCTURE

LAB PROGRAMS

23. Prim's Algorithm

```
#include <stdio.h>

#define V 4

int main() {
    int graph[V][V] = {
        {0,2,0,6},
        {2,0,3,8},
        {0,3,0,0},
        {6,8,0,0}
    };

    int selected[V]={0};
    selected[0]=1;
    int edge=0;
    printf("Edges in MST:\n");
    while(edge < V-1) {
        int min=999,a=0,b=0;
        for(int i=0;i<V;i++) {
            if(selected[i]) {
                for(int j=0;j<V;j++) {
                    if(!selected[j] && graph[i][j]) {
                        if(min > graph[i][j]) {
                            min = graph[i][j];
                            a=i;
                            b=j;
                        }
                    }
                }
            }
        }
    }
}
```

```
        }  
    }  
}  
  
printf("%d - %d : %d\n",a,b,min);  
selected[b]=1;  
edge++;  
}  
return 0;  
}
```

Output

Edges in MST:

0 - 1 : 2

1 - 2 : 3

0 - 3 : 6

=== Code Execution Successful ===